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Operability and control in process intensification and modular design: Challenges and opportunities

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Abstract

In this article, the importance of considering operability and control criteria in the analysis and design of intensified and modular processes is discussed. We first analyze the impact on operability of key factors including: (i) degrees of freedom, (ii) process constraints, (iii) numbering up vs. scaling up, and (iv) dynamic/periodic operation. Comparative examples are presented to showcase the pros and cons in intensified/modular systems vs. their conventional counterparts from operability and control aspects. Then we look into metrics and tools to address these challenges such as: (i) flexibility analysis, (ii) operability-based design, and (iii) advanced model-based control. Considering different conceptual design stages as synthesis intensification, steady-state design, and dynamic operational optimization, we highlight the need to incorporate different levels of operability considerations. Future research opportunities and perspectives are also identified, particularly emphasizing the importance of a holistic strategy for integrated design, operability, and control of intensified and modular process systems.

KEYWORDS
control, design and control, modular design, operability, process intensification

1 | INTRODUCTION

Facing a highly competitive global market with increasing awareness on environmental and safety issues, chemical production is making its way toward a paradigm shift to more efficient, more environmentally friendly, and more versatile.[1,2] Process intensification and modular design are regarded as promising solutions to pursue this structural transformation, gaining significant recent impetus in the chemical/energy industry and the chemical engineering research community to accelerate the deployment of these novel technologies as a joint effort.[3,4]

Process intensification (PI) aims to boost process and energy efficiency, enhance process profitability and safety, while reducing waste and emissions by embracing innovative process designs, which may be out of the box of current industrial practice.[5,6] A wide range of PI technologies have been developed,[7] some of which are already successfully commercialized such as reverse flow reactor,[8] reactive distillation,[9] and divided wall column,[10] to name a few. We have summarized at least seven activities, which will result in intensified processes[11]: (i) combining multiple process tasks or equipment into a single unit (e.g., reactive distillation, membrane reactor), (ii) tight process integration (i.e., material or energy integration), (iii) use of novel multifunctional materials (e.g., ionic liquid, zeolite), (iv) miniaturization of process equipment (e.g., micro-reactor), (v) changing of operation modes (e.g., pressure swing adsorption), (vi) application of enhanced driving forces (e.g., rotating packed bed), and (vii) advanced operational strategies (e.g., dynamic operation via advanced control).

On the other hand, modular design is a different while often concurrent concept to process intensification. It aims to dramatically reduce the size of process units to change from "the economy of scale" to small, distributed, and standardized plants with better flexibility and faster response to changes in demand, especially for utilization of unconventional feedstocks and for specialty chemical production.[12] A key question for modular design is the gain or loss in cost efficiency vs. design/operation agileness when comparing "numbering up" against conventional "scaling up." For many intensified technologies (e.g., micro-reactors, membrane reactors, alternative energy sources) which inherently function the most effectively at

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TABLE 1 Latest review/perspective articles on process intensification and modular design—an indicative list

References	Highlight	Year
13	Experimentation, modeling, and dynamic control of reactive dividing wall columns	2018
14	Microwave- and ultrasound-enhanced biodiesel production and role of catalysts/solvents	2018
15	Biofuel production from organic wastes with PI technologies	2018
16	PI technologies and application opportunities in water industry	2018
7	PI definitions and state-of-the-art technology development	2018
12	Status, challenges, and opportunities for modular manufacturing processes	2018
17	Microreactor fabrication, design, and applications	2018
11	PSE approaches for PI design, synthesis, and operability analysis	2018
18	Process intensification via dynamic/periodic operation	2018
19	PSE approaches for design and synthesis of intensified reaction and/or separation systems	2018
20	Reactive distillation and its novel technology variants	2018
21	Distributed optimization, control, and monitoring of large-scale PI systems	2019
22	Optimal operation, control, and scheduling of PI processes	2019
23	Process synthesis and optimization approaches for PI design and retrofit	2019
24	Optimization-based tools for PI in gas-to-liquids processes	2019
25	Challenges and opportunities for synthesis of operable PI systems	2019
26	Systematic approaches and procedures to generate, assess, and analyze PI solutions	2019
3	Activities under the RAPID institute to accelerate modular chemical process intensification	2019
27	Role of simulation and digitalization tools to assist modular process intensification	2019
28	Thermally coupled distillation processes for multicomponent separation	2019
29	Role of (advanced) materials in PI	2019
30	PI activities at ExxonMobil to address the dual challenge of energy and environment	2019
31	Model-based operability theory, applications, and software tools for modular PI systems	2019
32	Perspectives on computer-aided process intensification	2020
33	Equipment, material, and process innovation for carbon capture and conversion	2020

smaller scales, the combination of PI technologies with modular design may provide an encouraging synergistic process solution.[3]

Many excellent reviews have been offered on PI and/or modular design to elucidate the history, evolution, and state-of-the-art development of relevant technologies and applications. In Table 1, we provide a summary of the latest review/perspective papers to highlight the major scientific and industrial interests in this field since 2018. As can be noted, a diversity of intensified and modular technologies are being investigated to address the dual challenge of energy productivity/efficiency and environmental sustainability,[30] for example, reactive distillation,[24] intensified reactors,[15] alternative energy sources,[14] and advanced materials[33]. Another rapidly emerging research theme is computer-aided process intensification.[11,32] Early PI breakthroughs mostly relied on Edisonian efforts via experimentation.[34] The advanced computational methods and tools, with expertise of the process systems engineering (PSE) community, aim to support quantitative decision making by providing techno-economic evaluations as well as predictive capabilities on the design,[25,26] optimization,[19,23,24] and operation[18,21,22] of PI and modular systems.

Current efforts toward process intensification and modularization through PSE approaches can be generally classified as the following:

- Design and optimization
  - Process design, synthesis, and intensification — especially highlighting the phenomena-based approaches to generate innovative process solutions[35-37]
  - Optimization of modular production — to compare the efficiency, economics, and/or reliability of numbering up vs. scaling up[12,38,39]
- Optimal operation and control
  - Operability analysis — including steady state and dynamic feasibility, flexibility, controllability analysis, and so on[31,40]
  - Advanced model-based control — especially highlighting the need for distributed and decentralized model-based control in intensified and modular systems, as well as integrated considerations of design, control (and scheduling)[22,41]
  - Dynamic/periodic operation — to purposefully perturb operation along an optimal trajectory by exploiting process nonlinearity and output multiplicity[42,43]
  - Supply chain optimization and operational planning of modular production — especially highlighting the flexibility and mobility of modular processing[44,45]

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We focus in this article on the operability and control of PI and modular designs, which are essential requirements for any process, but even more important for intensified and modular designs.[46,47] The main contributions of this work are threefold. First, we highlight the need to develop a formal theory and fundamental understanding of operability and control in process intensification and modular designs. To this end, some key open questions will be discussed later in this paper using rigorous model-based analyses, which include:

i. How does the loss of degrees of freedom (DOFs) affect the operation of an intensified process compared to its conventional process counterpart?
ii. Operability concerns result from the violation of inequality process constraints during actual operation (under uncertainty and disturbances). For the role of process constraints, what is the difference between intensified vs. conventional processes?
iii. While PI is affected by the loss of DOFs, the numbering up of modular designs can contribute to additional DOFs. What is the trade-off between economics and operability/control in an intensified modular production process?
iv. Inspired by the periodic processes, is it possible to operate the process in a dynamic manner following an optimal trajectory (such as to minimize energy consumption)?

Second, we emphasize the need to integrate operability and control analysis at the conceptual PI design stage, leveraging state-of-the-art PSE approaches for process synthesis intensification, design under uncertainty, and advanced model-based control optimization. The design of a chemical process determines its inherent operability dictating how well the process rejects disturbances, how severely the multiple variables interact, and how easily and smoothly the system moves from one operating condition to another.[48] As will be shown later, the impact of design on operability and control becomes more profound in intensified process systems. At each design stage, different approaches are used to describe the corresponding design space such as dynamic high-fidelity modeling, steady-state design, and superstructure-based synthesis (e.g., using phenomena-based bottom-up representation). The integration of proper operability analysis approaches with different synthesis/design approaches is therefore necessitated to systematically generate intensified/modular designs with guaranteed and consistent operability performance.

Finally, perspectives are discussed toward a unified strategy and software platform to integrate design, operability, and control of intensified and modular process systems. A major gap to accelerate the development and deployment of these novel technologies in chemical process industry is the lack of widely used commercial software (or even a software prototype), which can fully support systematic PI and modular design optimization with operational analysis. Driven by several recent national initiatives such as the Department of Energy RAPID Institute[3] and IDAES Project,[49] attempts have been initiated from academia to develop software prototypes for computer-aided process design, synthesis, and intensification. We will review the advances in these academic tool developments and the opportunities to synergize with ongoing industrial efforts on process modeling, simulation, and digitalization.

The remainder of this article is organized as follows: Section 2 rigorously analyzes the above operability and control challenges in intensified and modular designs using an illustrative example to compare a reactive distillation process and a reactor-distillation-recycle process. Section 3 looks into the opportunities to address these challenges by extending PSE approaches for systematic process intensification and modularization with advanced model-based control and operability analysis. Section 4 gives the concluding remarks.

2 | OPERABILITY AND CONTROL CHALLENGES IN INTENSIFIED AND MODULAR SYSTEMS

The inherent process physics and dynamics in PI and modular systems can be very different from those in well-established conventional processes, normally posing more demanding requirements on operation and control. For example, a key difficulty with process intensification is that it can result in tight integration of tasks with less DOFs, narrower operating windows, and typically faster process dynamics. In this section, we discuss questions pertaining to the following topics: (i) loss of DOFs, (ii) role of process constraints, (iii) numbering up vs. scaling up, and (iv) role of dynamic/periodicity.

2.1 | Loss of DOF

The loss of DOFs occurs mostly in PI systems, which combine multiple process steps into a single unit or with tight mass/energy integration between units. Note that the basis of comparison is at the process level to design an intensified process or its conventional counterpart process to complete a given production task, instead of comparing a piece of intensified equipment vs. a conventional one. To showcase how the DOFs are impacted by process intensification, we give an illustrative example comparing an intensified reactive distillation (RD) process against a conventional reactor-distillation-recycle process as shown in Figure 1.

2.1.1 | DOF analysis

To investigate how the DOFs change in these two processes at different process design stages, we first perform a general DOF analysis using three types of models: (i) high fidelity dynamic modeling, (ii) steady-state modeling, and (iii) superstructure-based synthesis modeling. The detailed DOF analysis results are presented in the Supporting Information Appendix A. Regardless of model types, RD consistently has less DOFs compared to its conventional counterpart. Some general rules can also be summarized which result in the loss of DOFs in such a task-integrated system as reactive distillation:

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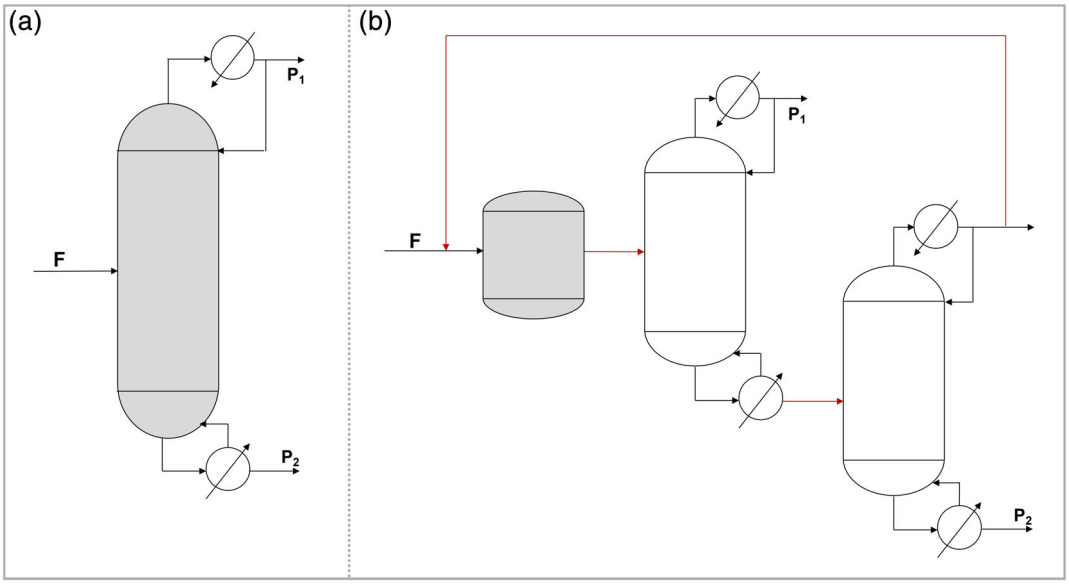


FIGURE 1 Degrees of freedom comparison: (A) reactive distillation, (B) reactor-distillation-recycle [Color figure can be viewed at wileyonlinelibrary.com]

- The coupling of reaction and separation in a single unit requires reaction and separation to take place under the same temperature and pressure conditions. This mostly affects the thermodynamic DOFs and results in the concern of a reduced operating window.[20]
- The full integration of the reactor and the distillation in a single RD unit, at the same time, converts the interconnecting streams between units (e.g., reactor outlet stream, recycle stream) to internal flows within the unit. Thus, the potential measurement and manipulation of these steams (e.g., through valve) are no longer available.
- The reduction in the number of units in RD results in the loss of thermodynamic, design, and operational DOFs.

In addition to these empirical observations, it is worth highlighting a theoretical development from Baldea.[34] The author rigorously proved that, an integrated process, which consists of *n* units in series with identical material holdup and an infinitely large material recycle stream connecting the last and the first units, is statically equivalent to an intensified process consisting of a single unit with the same material up as each integrated unit. However, the dynamics of the intensified process is *n* times faster than the integrated one, while losing available manipulated variables for control in the streams between each unit.

2.1.2 | A numerical case study

To compare the open-loop and closed-loop dynamic behaviors, in what follows we will present a comparative case study for olefin metathesis to produce 2-butene and 3-hexene from 2-pentene. Two design alternatives are considered, namely an intensified reactive distillation process and a conventional reactor-distillation-recycle process as depicted in Figure 1. The detailed design and operation parameters can be found in the Supporting Information Appendix B.

Considering a fixed design configuration, the reactive distillation column has two available operational DOFs, that is, reflux ratio and distillate flowrate (Table A1). On the other hand, six operational DOFs are available for the control of the reactor-distillation-recycle process: (i) reactor outlet flowrate controlled via valve stem position, (ii) distillate flowrate and reflux ratio for distillation column 1, (iii) reflux ratio and boilup ratio for column 2, and (iv) recycle ratio. We first perform a series of open-loop analyses on each of the above DOFs (or manipulated variables). The dynamic responses in RD and reactor-distillation-recycle are illustrated in Figure 2.

Two differences can be summarized based on Figure 2: (i) the dynamic response of RD is much faster than that of the reactor-distillation-recycle, and (ii) the reactor-distillation, with significant material recycling, shows a typical two-time-scale behavior in such integrated systems. For example, the response of bottom product purity to a step change in the valve stem position (Figure 2(B)) clearly exhibits an initial fast transient followed by slow dynamics. However, this dynamic behavior is not observed in the RD process.

For the RD closed-loop control, two PI controllers are designed in which reflux ratio and distillate rate are used as manipulated variables to respectively control the top and bottom product purity as two single-input single-output (SISO) systems. As shown in Figure 3, PI control can meet the requirement of set point tracking to RD top and bottom product purity.

Among the six DOFs for reactor-distillation-recycle, two pairing schemes are selected and tested to control the top and bottom product purity with SISO PI controllers. The results are shown in Figure 4. It can be noted that PI control can also perform satisfactorily for set

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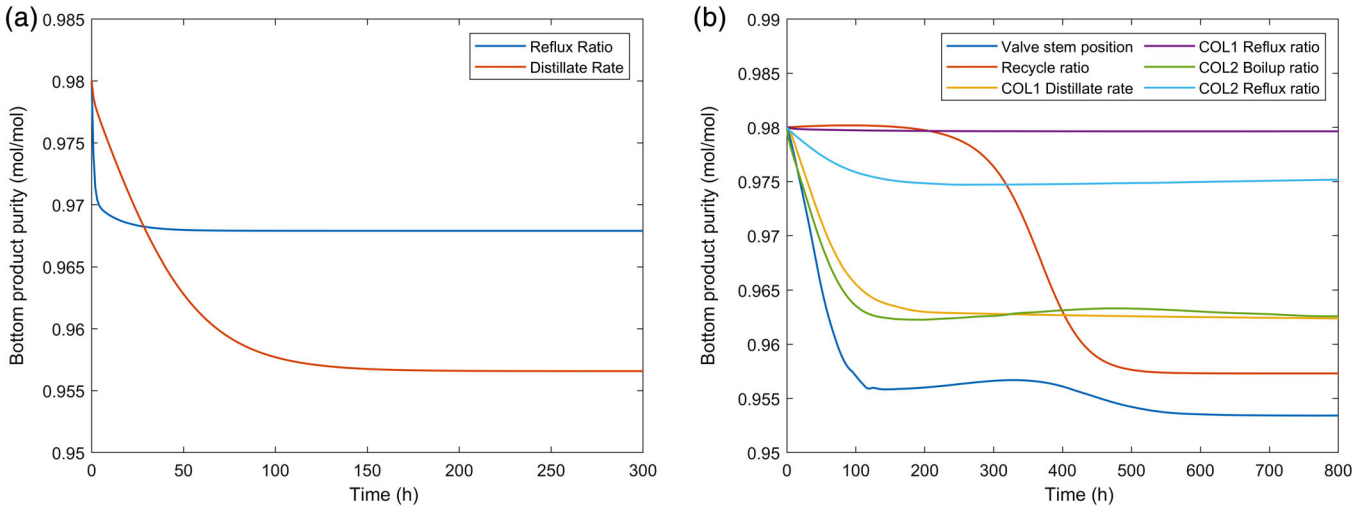


FIGURE 2 Open-loop step response: (A) reactive distillation, (B) reactor-distillation-recycle [Color figure can be viewed at wileyonlinelibrary.com]

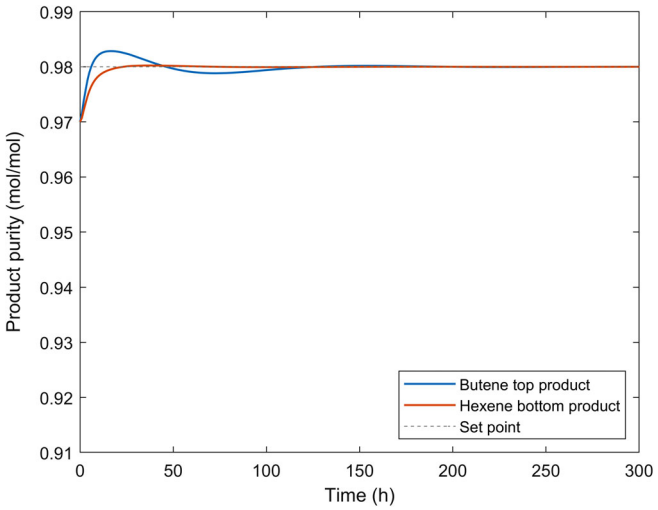


FIGURE 3 PI control closed-loop performance for set point tracking in RD column [Color figure can be viewed at wileyonlinelibrary.com]

point tracking in this conventional process but only when a good pairing scheme is selected.

2.2 | Role of process constraints

From a model-based point of view, any operability, safety, or control concerns can be viewed as the violation of process modeling constraints — typically the inequality ones which define product specifications, safe operation region, design capacity, and so on. These violations can be caused by the existence of disturbances and uncertainties, changes in operating conditions, product specifications, and so on.[50,51] In this section, by analyzing the role of constraints, we compare the operability performance of the reactive distillation process and reactor-distillation-recycle process as a motivating example.

2.2.1 | Temperature and pressure bounds

The combination of reaction and separation into a single unit restricts these two tasks taking place under the same temperature and pressure conditions. In this context, the operating window of RD is conceptually the *intersection* of the reaction operating window and the distillation operating window.[20,41] However, the operating window of reactor-distillation-recycle is ideally the *union* of the operating windows of reaction and distillation.

Recall the olefin metathesis case study introduced in Section 2.1.2 — the reaction occurs at atmospheric pressure between 270.15 K and 340.15 K, which results in a rather limited operating window in the space of temperature–pressure (T–P). Figure 5 depicts the T–P window for reaction, distillation, and reactive distillation by sampling through the design and operation space for the specific production task. It can be observed that, in this example, RD is faced with a significant reduction of operating window as well as much stricter design and operation constraints, compared to the much more flexible operating window enabled by distillation process.

2.2.2 | Flowrate bounds

For the stream flows connecting different units in a process flowsheet, flowrate bounds are normally given either to avoid unreasonably large flows (e.g., for recycle stream) or to ensure feasible equipment design (e.g., pipe). A notable difference of reactive distillation, compared to the flowsheet of a reactor followed by a train of distillation columns, is the conversion of external connecting streams (e.g., reactor outlet stream, recycle stream) to internal flows within the

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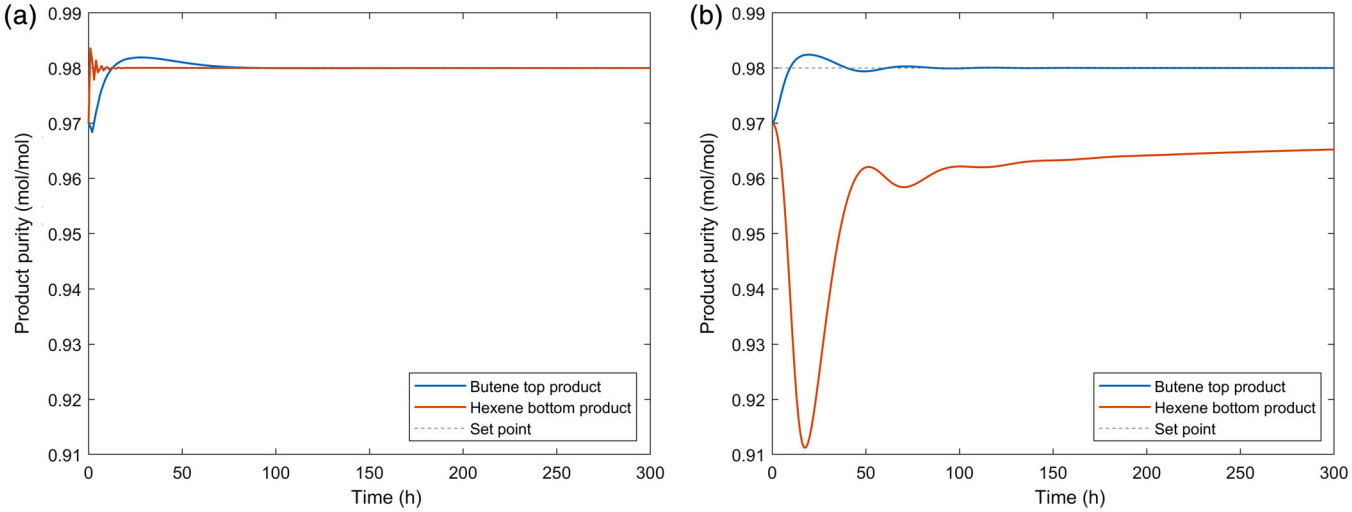


FIGURE 4 PI control closed-loop performance for set point tracking in reactor-distillation-recycle process: (A) Manipulated variables—COL1 distillate flowrate and COL2 boilup ratio, (B) Manipulated variables—COL1 distillate flowrate and flowsheet recycle ratio [Color figure can be viewed at wileyonlinelibrary.com]

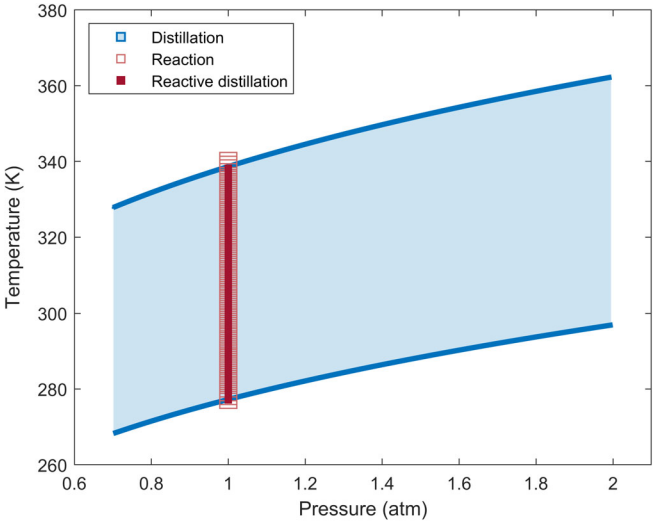


FIGURE 5 Temperature–pressure operating window for olefin metathesis [Color figure can be viewed at wileyonlinelibrary.com]

unit — which emphasizes the impact of design parameters on flowrate bounds. Column diameter has been found to be the most limiting design parameter for distillation operation under uncertainty. A given column diameter can only accommodate a certain range of flowrate uncertainties, beyond which will cause flooding issues. Using the example of olefin metathesis, we compare the impact of column diameter on an allowable feed flowrate range in the RD column and the reactor-distillation-recycle process.

For a given reactive distillation column diameter (or the diameter of Column 1 or 2 in the reactor-distillation-recycle process), we first characterize the "steady-state feasible region" in which the process is feasible for operation and can satisfy the top and bottom product purity specifications under uncertainties in feed flowrate. Then, the "dynamic feasible region" is determined where the process is required to meet process specifications with PI controller on under time-variant feed flowrate uncertainties. Conceptually, the dynamic feasible region should be a subset of the steady-state region since the latter assumes perfect control. Several observations can be made based on the results depicted in Figure 6:

- RD can accommodate a larger set of feed flowrate uncertainties than its conventional counterpart at the same over-design factor (i.e., over-design compared to the minimum required column diameter at nominal feed flowrate 100 kmol/h). This is because the conventional process has around twice larger external and internal flowrates than RD.
- The reactor-distillation-recycle process reaches a maximum flowrate at 142 kmol/h regardless of the increase of column diameter. This is due to the fact that a prespecified upper bound is met for recycle flowrate at 425 kmol/h compared to the nominal at 300 kmol/h.
- The dynamic feasible region of RD has a notable mismatch with its steady-state feasible region, which can be up to 11 kmol/h. This is due to the more complex interaction between process variables in RD, which makes the process more vulnerable to constraint violation under uncertainties and/or control actions. This emphasizes the need of simultaneous dynamic design and control under uncertainties in such intensified systems.

2.2.3 | Process specifications

Given a step change in feed flowrate (+1 kmol/h), Figure 7 shows that RD has a larger deviation from the purity specifications than that in the reactor-distillation-recycle process.

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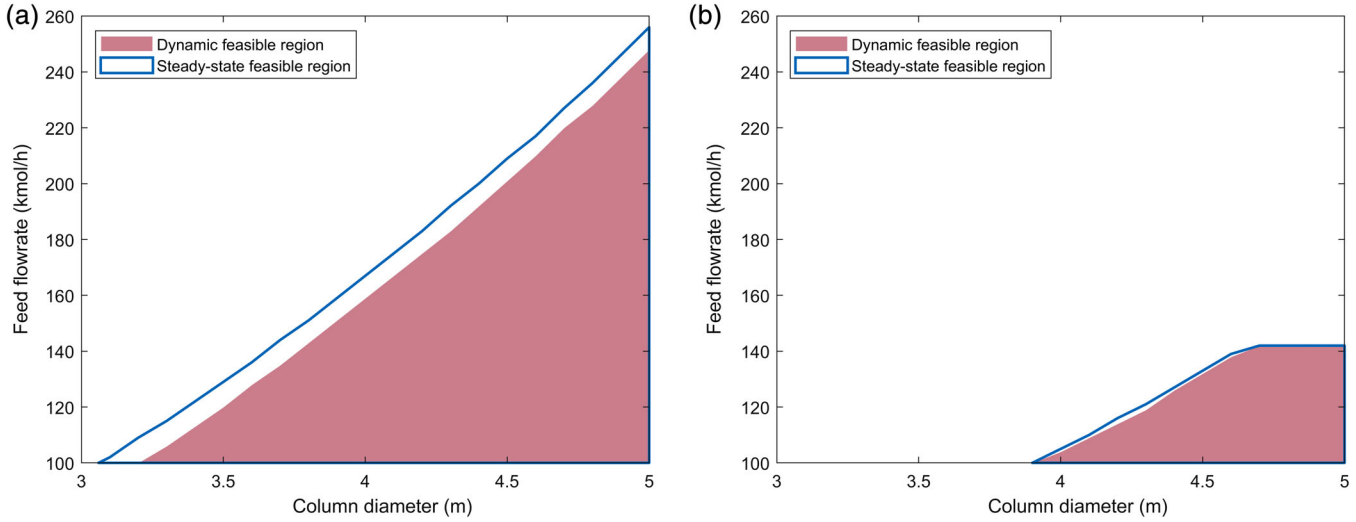


FIGURE 6 The impact of column diameter on flowrate constrains [Color figure can be viewed at wileyonlinelibrary.com]

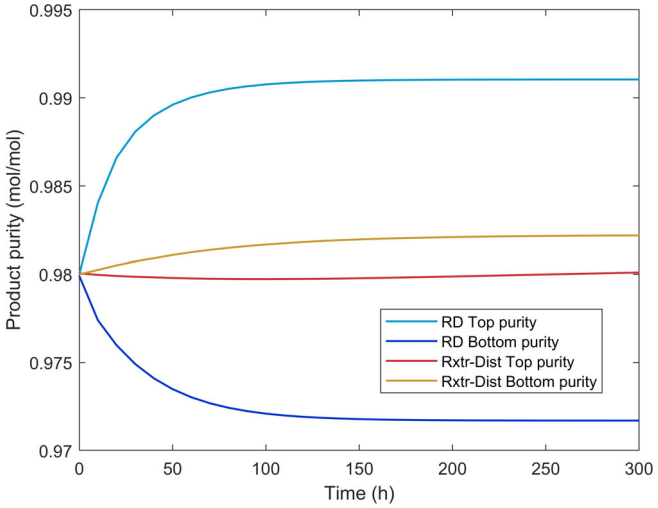


FIGURE 7 Open-loop response to feed flowrate step change [Color figure can be viewed at wileyonlinelibrary.com]

To test the closed-loop performance of these two processes, a sinusoidal disturbance is considered in the feed flowrate with a period of 24 h and an amplitude of 8 kmol/h. With the PI controllers on, the control output profiles for RD and reactor-distillation-recycle are given in Figure 8. While the conventional process is well controlled, the RD process has notable off-specifications.

Given the insufficiency of the SISO PI controllers for RD disturbance rejection, another explicit/multi-parametric model predictive controller (mp-MPC) is designed following the PAROC frameowrk,[52] treating the RD column as a multi-input multi-output (MIMO) process system. The closed-loop performance of both control schemes is shown in Figure 9, indicating better control performance with the mp-MPC controller in terms of meeting product purity specifications under disturbance.

2.3 | Numbering up vs. scaling up

In this section, we investigate the impact of modularization on operability and control. We focus on the question if/how the previously mentioned operability and control concerns of an intensified unit can be addressed by introducing extra modular and parallel units (at the expense of more investment cost). Another potential benefit of using modular and parallel units is to enhance process reliability which indicates the probability of a system to perform its designated function over a specified time interval. While this topic is beyond the scope of the current paper, we refer the readers to the recent works[53,54] for more information.

Continuing with the example on olefin metathesis, an alternative flowsheet is constructed against the single RD unit which consists of two modular RD units operated in parallel as shown in Figure 10. From economics point of view, the modular flowsheet features a total annualized cost 18.8% higher than that of the single RD unit.

Recall the discussion in Section 2.1 on loss of DOFs due to process intensification, modularization on the other hand provides the opportunity to increase DOFs because of the increased number of process units (including also auxiliary units such as mixers and splitters) and interconnecting streams. To test the control performance for rejecting the sinusoidal feed flowrate disturbance in Section 2.2.3, PI controllers are designed to control each modular RD unit as per the pairing scheme 1 given in Table 2. If the two modular RD units adopt the same controller designs, they will function synchronously showing a closed-loop performance similar to the original single RD unit. However, the overall process closed-loop performance can be improved in this example by selecting different pairs of control inputs and outputs as depicted in Figure 11(A). A second pairing scheme, as detailed in Table 2, can be designed to control units at a process level to ensure that the final products are on specification without monitoring product specifications from individual units. The additional DOFs resulted by modularization can again contribute to a better disturbance

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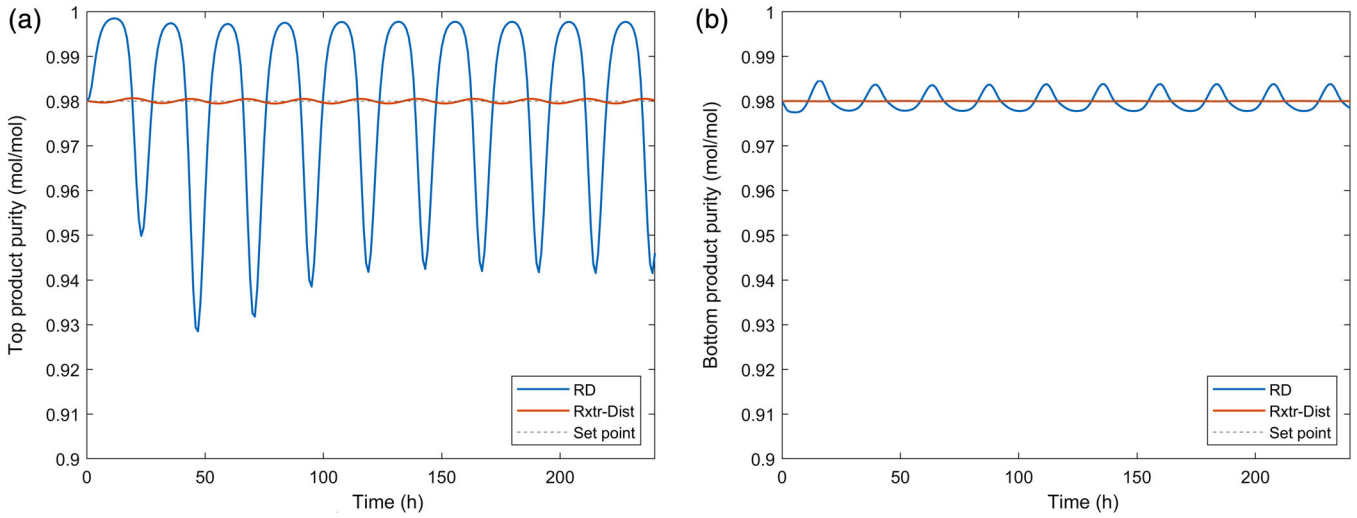


FIGURE 8 PI control closed-loop performance for disturbance rejection in RD and reactor-distillation-recycle (Rxtr-Dist): (A) Top butene product purity profile, (B) Bottom hexene product purity profile [Color figure can be viewed at wileyonlinelibrary.com]

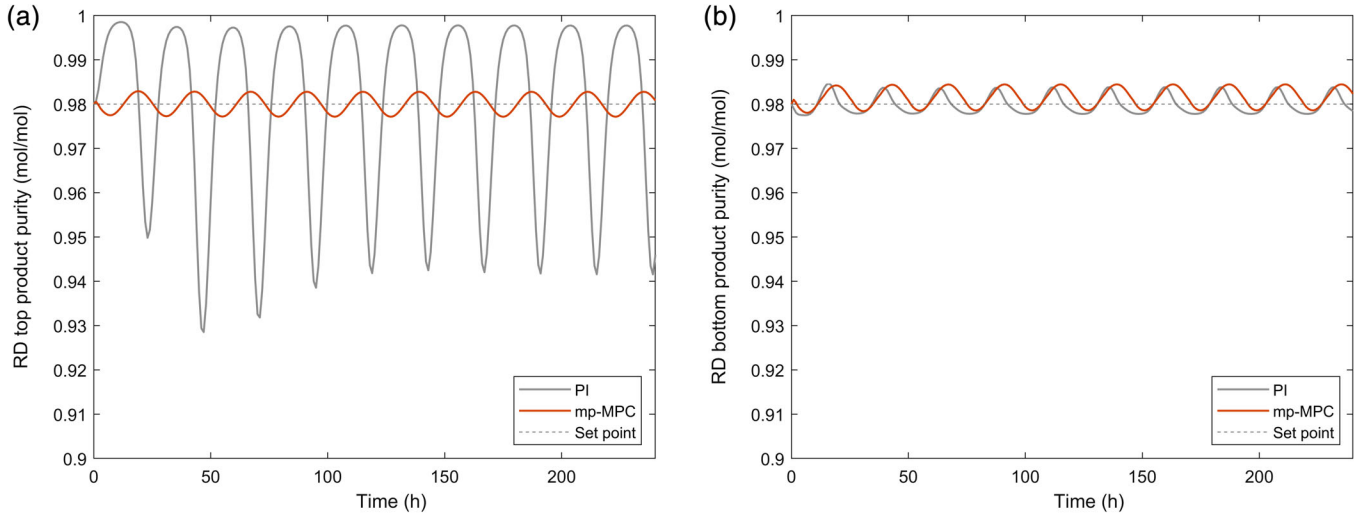


FIGURE 9 PI and mp-MPC control for RD disturbance rejection: (A) Top butene product purity profile, (B) bottom hexene product purity profile [Color figure can be viewed at wileyonlinelibrary.com]

rejection performance in terms of the final product purity as shown in Figure 11(B). This pairing scheme also brings the benefit of one remainder DOF which can be used to achieve other operational objectives of interest, such as to minimize energy consumption. Note that under both pairing schemes, the two modular RD units are operated along different trajectories. This observation indicates further opportunities to explore the design and control optimization of this modular flowsheet by simultaneously considering modular unit design, control structure, and closed-loop performance.

Some other questions or challenges in modularization include:

- As indicated above, the operational flexibility can be improved with a number of modular designs by enabling synchronized operation or tailored operation for individual units. Thus, it opens up questions on how to optimize the operation/control strategies for each of the units and how to perform optimal decision-making when the units need to alter operation states in response to changes in production plan.
- The trade-off between the number of control variables vs. control efficiency needs to be addressed when the units are numbering up. For example, how to ensure a certain temperature profile across the parallel modular units? In addition to the unit-by-unit control analogous to controlling a single-unit operation, indirect control systems can be a good option to reduce the control variables as shown in Hasebe[55] for microreactors.
- Distributed decision making is also necessitated which decomposes the large-scale modular intensified process into constituent subsystems with corresponding localized decision-making agents.[21,56] The control decisions are then coordinated for the overall process following approaches such as distributed model predictive control.[57]

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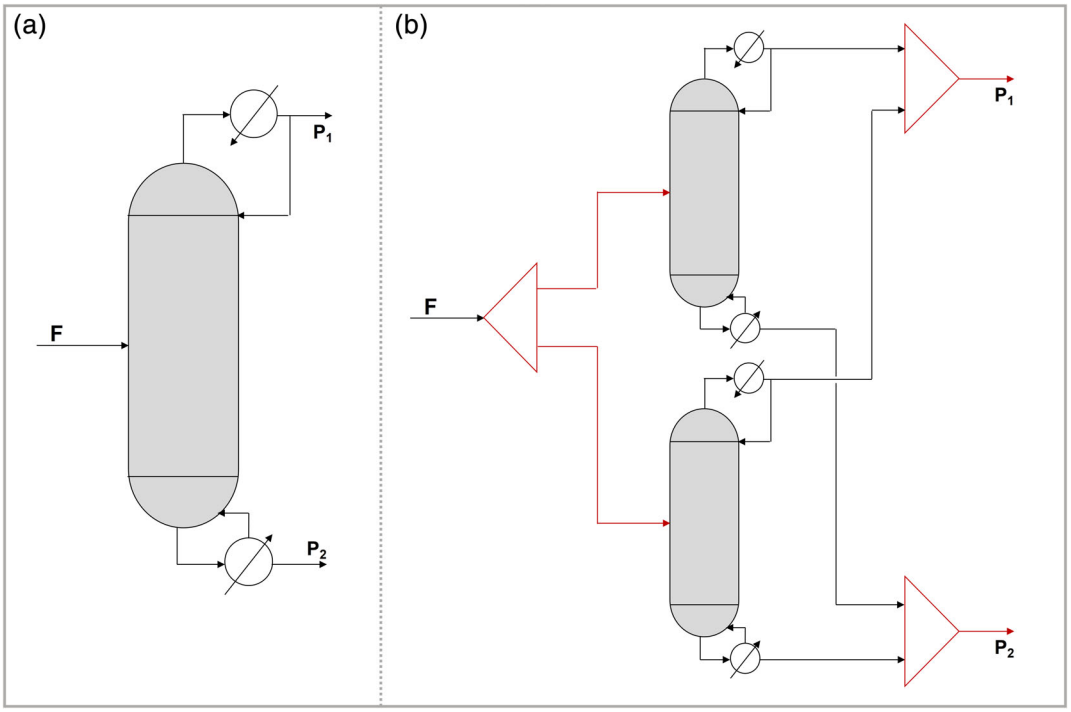


FIGURE 10 Reactive distillation for olefin metathesis: (A) A single unit and (B) two modular units [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 2 PI control pairing schemes for modular RD units

	Pairing scheme 1		Pairing scheme 2	
	Input	Output	Input	Output
Controller 1	Feed split ratio	RD1 top product purity	Feed split ratio	Top product purity after mixing
Controller 2	RD1 distillate rate	RD1 bottom product purity	RD2 reflux ratio	Bottom product purity after mixing
Controller 3	RD2 distillate rate	RD2 top product purity	RD2 distillate rate	RD2 top product purity
Controller 4	RD2 reflux ratio	RD2 bottom product purity		

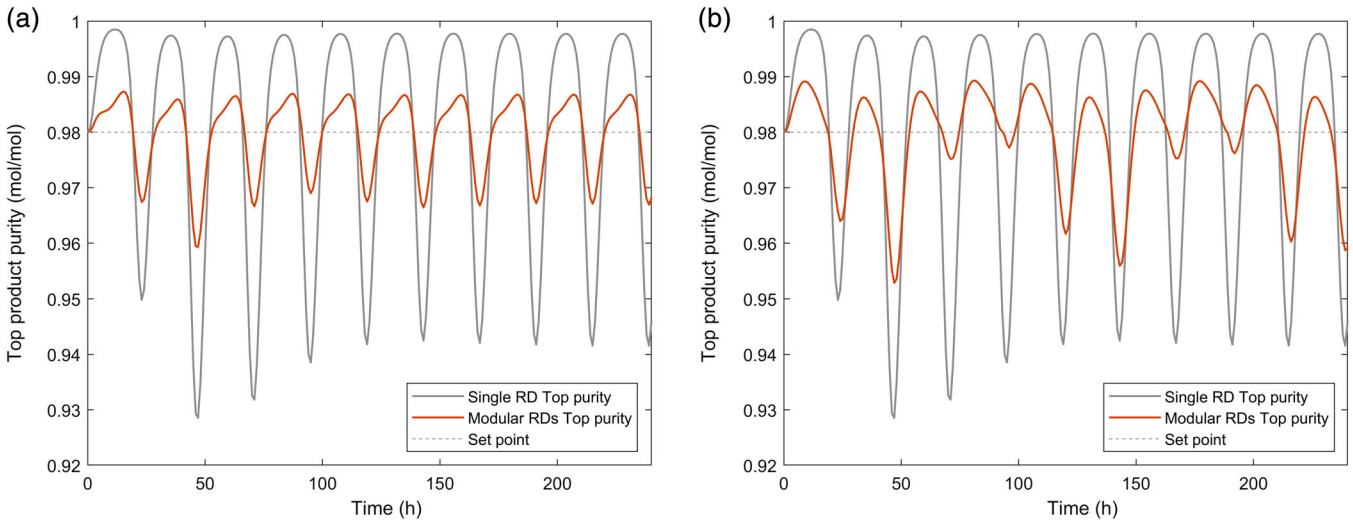


FIGURE 11 PI control closed-loop performance for modular RD units vs. single RD: (A) pairing scheme 1, (B) pairing scheme 2 [Color figure can be viewed at wileyonlinelibrary.com]

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- On the plant-wide level, another issue is the potential time delay caused by extra mixing of the products from each modular unit, which needs to be taken into account for controller design.

2.4 | Role of dynamic/periodicity

Dynamic/periodic operation plays a key role in the following intensified/modular systems:

2.4.1 | Switching from batch to (semi-)continuous production

There has been increasing interest in developing continuous production processes instead of the traditional batch processes, especially in the fine chemical and pharmaceutical industry.[58,59] Continuous processing offers the advantages of higher product throughput and yield, easier scaling up, more consistent product quality, safer operation under extreme conditions, and so on. A key challenge for the design and control of the new continuous processing scheme is to meet the strict pharmaceutical product regulations. As such, Su et al.[60] introduced a concentration control strategy to operate an existing stirred tank batch crystallizer under continuous mixed-suspension mixed-product removal conditions, the start-up procedures and on-line control conditions of which stay within the design-space of the original batch operation. For another semi-continuous multicolumn countercurrent solvent gradient purification process for the production of monoclonal antibodies, model predictive control strategies were explored to maximize product purity and recovery yield under disturbances (given the fact that in practice these systems are operated at sub-optimal conditions to guarantee robustness).[61,62] To answer a more fundamental question of when it is preferred to replace an existing batch reactor with a continuous flow reactor, Costandy et al.[63] proposed a methodology to assess the feasibility and optimality of this transition by matching the operation trajectory of these two systems.

2.4.2 | Inherently periodic systems

The periodically operated systems offer the advantage from the PI view point to integrate multiple tasks into a single unit at different time scales. A number of periodic systems have been developed and employed in chemical process industry. For example, cyclic distillation can be retrofitted from traditional distillation columns by changing the internals and operation mode but to offer higher tray efficiencies and lower energy consumptions as already been demonstrated in industrial applications.[64] Other technologies, such as reverse flow reactor, simulated moving bed, and pressure swing adsorption, also provide the benefits of standardization and parallelization by using modular process equipment.[18] However, operational challenges result from: (i) the lack of a true steady state but only attaining a pseudo (or cyclic) steady state, (ii) the design and coordination of switching strategies, and (iii) the requirement of simultaneous design, operability, and control due to the intrinsic interlink of these aspects in periodic systems.

Taking the pressure swing adsorption (PSA) as an example, this category of periodic process is mostly described with partial differential equations (PDE) in space and time, or equivalently in nonlinear programming models via full discretization.[65] Thus, specialized dynamic simulation/optimization algorithms become another key enabler in addition to modeling the inherent physical complexity. Only a few work are available for the control of the PSA systems, among which model predictive control is mostly applied and suggested. In the work by Khajuria and Pistikopoulos,[43] explicit/multi-parametric MPC controller outperformed the conventional PID controller in terms of faster control response with less control effort for feed flowrate disturbance rejection in a 4-bed PSA process to separate a gas mixture of hydrogen and methane. To afford computationally tractable on-line/off-line MPC control, reliable model reduction is required.

2.4.3 | "Purposeful" dynamic/periodic operation

Periodic perturbation can be introduced to chemical systems to achieve better process performance than operating at their optimal steady states. A representative example is the periodic operation of conventional chemical reactors by oscillating pressures, concentrations, or temperatures to shift process over reaction equilibrium.[66,67] A more recent work by Yan et al.[42] reduced the energy consumption of a binary distillation system by periodically altering column operation among multiple steady states while maintaining the time-average product concentration at purity specification. Dynamic PI is achieved based on the observation that the target product can be obtained as a blend of two auxiliary products (one of higher purity than desired, the other of lower purity), both of which have lower specific energy consumption. However, open questions remain on: (i) how to optimize the operation trajectory, and (ii) how to control these systems to account for disturbances and slow drifts.

Regarding the operation trajectory, Freund and Sundmacher[68,69] developed a methodology to design intensified chemical reactors based on the concept of elementary process functions. The main idea was to track a fluid element on its way through the reactor and to optimize the fluxes (e.g., heat flux, reaction flux) along its way to enforce optimal state space profiles in space and time with detailed considerations of reaction kinetics, thermodynamics, transport kinetics, and so on. Nikačević et al.[41] further proposed a three-step hierarchical framework to fully integrate process intensification, operation, and control based on dynamic optimization. Namely: Step 1 — Process intensification by optimizing fluxes and driving forces, Step 2 — Controllability study via robust optimization, and Step 3 — Controller design and tuning. Two key questions result from the above framework which also open up more opportunities toward a holistic framework for process intensification and operation:

- *Definition of driving forces to define the optimal operation trajectory* — Various driving forces have been defined pertaining to specific

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types of processes. For example, the previously mentioned elementary process function for reactors,[69] attainable region theory to determine ultimate performance bounds in reaction/separation systems,[70,71] reactive driving force for (reactive) distillation systems,[72,73] Gibbs free energy-based driving force constraints for reaction/separation systems,[74] and so on. Given the large scope of PI and modular designs, a well-defined driving force formulation is clearly needed, which may not suggest directly the type of equipment but can rapidly screen the design space and provide preliminary design and operation information (including temporal and spatial profiles).

- *Mathematical formulations* — While dynamic modeling and optimization provide accurate descriptions of process systems in both temporal and spatial domains, it is highly demanding (and currently limited) in solution algorithms, problem scales, and computational complexity. Instead of applying dynamic optimization all the way through design, operation, and control, an alternative way is to leverage state-of-the-art steady-state synthesis and operability analysis methods to reduce the design space before employing dynamic optimization to close the loop for design and control optimization as proposed in the study by Tian et al.[75] However, a missing link in the current steady-state synthesis methods[36,37,76] lies in the lack of temporal considerations — which is particularly important to discover the dynamic PI opportunities.

2.4.4 | Dynamic operation driven by demand response

Compared to traditional steady-state operation, dynamic operation can also be advantageous when exogenous uncertainties exist such as fluctuating energy pricing, product demand, raw material availability, daily/seasonal disturbances, and so on. A promising approach to tackle this problem is via the integrated scheduling and control leveraging dynamic optimization formulations. A representative application is the air separation units (ASU) under real-time electricity pricing and customer demands, which has been investigated by a number of works.[77-79] For example, Dias et al.[79] proposed a framework for the integration of scheduling and MPC control, which consisted of: (i) an outer loop to integrate state-space process model, offline-MPC, and scheduling problem, and (ii) an inner loop to solve online MPC problems. Results of the study showed cost reductions up to 1.4% for the ASU operating under real-time energy pricing changes. To reduce the online computational load, Pistikopoulos and co-workers[80] introduced a unified framework for process design, scheduling, and control using multiparametric programming techniques to derive offline maps of optimal receding horizon policies for: (i) explicit model predictive control laws as a function of design and scheduling actions and (ii) explicit scheduling policies as a function of design and aware of closed loop dynamics. As the extension to intensified process systems, Otashu and Baldea[81] developed a demand response-oriented dynamic modeling and scheduling approach with application to a power-intensive membrane-based chlor-alkali process in response to electricity price changes in the short-term electricity market (i.e., 15 min). A 7.3% saving in energy cost has been reported by fluctuating current electrolysis density following electricity price signals as well as by depleting stored product to sustain demand during peak electricity price periods.

An outstanding challenge for demand response-oriented dynamic operation is posed by the inherently periodic systems such as PSA. The matching of periodic process dynamics with market dynamics needs to be calibrated based on the sophisticated process design and operation model as discussed in Section 2.4.2. Another gap for such operational optimization analysis in most PI systems lies in the lack of proper dynamic process models with sufficient accuracy and tractable computational requirement.

On the other hand, modular designs can play a critical role toward a flexible and agile production system in response to market uncertainties, the potential of which has not been fully investigated. With a number of modular and parallel units instead of a single unit, the operational flexibility of a plant can be increased by: (i) turning up/down modules, (ii) producing multiple products, (iii) using the modules for different functionalities, and so on. In this context, the start-up/shut-down dynamics and operation mode transient in the modular processing network need to be considered in the operational optimization formulation. A decision-making tool is also needed, which can evaluate the trade-off between the (extra) cost expenses and operational benefits due to modularization.

3 | RESEARCH OPPORTUNITIES AND PERSPECTIVES

In the previous sections, we have discussed several specific dynamic and operating characteristics in PI and modular designs and the resulting challenges from operability and control point of view. The following research opportunities are thereby identified and highlighted for intensified and/or modular systems (although also important for conventional processes):

- Theoretical developments are necessitated to understand the process dynamics, operability, and control for a wider range of modular and intensified systems (e.g., task-integrated systems, micro-reaction systems, rotating equipment).
- Conceptual design and operational optimization approaches for modular processing systems are needed to systematically analyze the trade-off between module sizing, profitability, and operational flexibility.
- The integration of operability and control metrics at an early PI design stage will be beneficial, due to the tight interaction between design and operating parameters in such systems (as discussed in Section 2).
- Distributed decision-making structure and advanced model-based control will be key enabling tools to ensure the actual operational performances of an integrated process with multiple modular and/or intensified process equipment. Advanced model reduction

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techniques and dynamic optimization algorithms should also be developed in support of this goal.
- A holistic framework and software platform for computer-aided design, operability analysis, and control optimization of modular PI systems can be highly beneficial to accelerate the technology development and deployment in chemical process industry.

3.1 | Toward a formal theory for operability and control in intensification and modularization

In addition to the questions discussed on the loss of DOFs, role of process constraints, choice of scaling up or numbering up, and dynamic PI, several other key open questions include:

- When is an intensified and/or modular process solution desirable considering (the trade-off between) its economics potential and operational performance?
- How to systematically generate intensified and/or modular designs at conceptual design stage and to analyze their operability performance?
- How to define an operability index to quantify the operational performance of an intensified and/or modular system?
- How will the optimal design decision, as well as the extent of process integration, change with respect to the above operability index?
- What is the role of advanced materials in terms of operability and control?
- How to define the optimal dynamic operation trajectory?
- When is it preferable to operate such systems at equilibrium and when at non-equilibrium?
- How to incorporate operability, control, safety considerations together with the design of PI and modular systems?

3.2 | Conceptual design of modular processing systems with enhanced operational flexibility and multifunctionality

For the design of process intensified systems, to come up with potentially unknown innovative process solutions, phenomena-based design approaches[25,26,32] have been advocated to represent chemical processes utilizing physicochemical building blocks (e.g., mass transfer, heat transfer) rather than equipment/flowsheet configurations. Examples of using such methods (e.g., Generalized Modular Representation Framework, GMF) to represent intensified reactive distillation and conventional reactor-distillation-recycle processes can be found in the Supporting Information Appendix A and Figure A.2. However, modular design options have not yet been considered in these approaches to evaluate if and how modularization can bring in economic and/or operational benefits in tandem with intensification. In light of this, we take the GMF[35,74] as an example and briefly discuss how it can be extended to represent modular design flowsheet.

As shown in Figure 12, GMF uses aggregated phenomenological modules (e.g., mass/heat exchange modules) to represent chemical processes. The mass and/or heat transfer extent in each module is governed by thermodynamic-based driving force constraints. The modular network can be designed at steady state but preferably in a dynamic manner to exploit the operational flexibility. Conceptually, these modules can be operated under the same conditions and connected in parallel, or in stack, as in Figure 12(A). Certain modules can be turned up or turned down during operation (Figure 12(B)). The connections between modules can be flexibly switched from producing a single product (Figure 12(A)) to producing multiple products (Figure 12(C)). Each mass/heat exchange module can be used specifically for different functionalities such as separation, reaction, or reactive separation as illustrated in Figure 12(D).

3.3 | Metrics for operability analysis

Operability describes in general the inherent ability of the plant to perform satisfactorily under conditions different from the nominal design conditions, which can include the requirement to adjust to changes in operating conditions, disturbances, product specification, and so on.[50,51] Extensive efforts have been made in the PSE community over the past several decades to quantify process operability performance from different aspects as summarized below (see detailed reviews[11,40,50,82]):

- Steady-state and dynamic flexibility analysis — to determine the maximum uncertainty/disturbance that can be handled at the nominal operating point in a fixed design configuration[83-85]
- Steady-state and dynamic controllability analysis — to determine the "best" dynamic performance achievable for a system under closed-loop control[48,51,86]
- Operability-based analysis — to determine whether a controller can perform its mission under the constraints posed by the existing input variables and desirable output variables using linear/nonlinear input/output mapping methods[31,87]

Given the operational concerns in PI and modular systems, operability analysis should be performed at a design stage as early as possible. The assessment can be either via posterior analysis after the process design is obtained or preferably integrated with design decision-making, the latter of which will be discussed with more detail later in Section 3.4. Several gaps need to be addressed to apply the above theoretical methodologies in PI systems:

- These model-based analysis approaches need to be tailored according to the unique operation challenges posed by intensification and modularization in tandem with the theory development in such processes.
- Taking the seminal flexibility analysis[50] as an example, it requires a rigorous and sophisticated mathematical reformulation of the process model incorporating Karush–Kuhn–Tucker optimality

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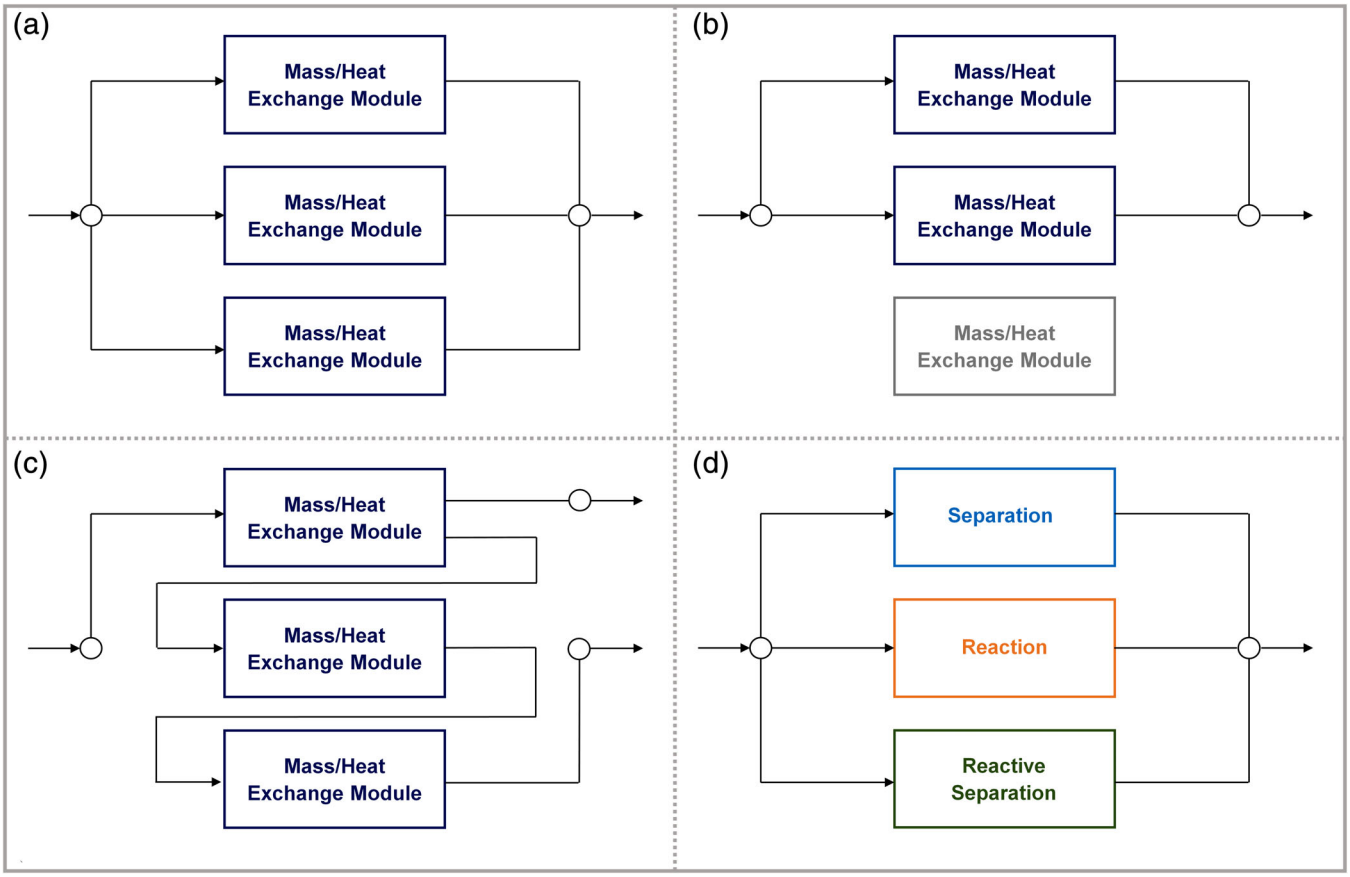


FIGURE 12 Conceptual modular design via Generalized Modular Representation Framework: (A) parallel operation, (B) flexibility for turning up/down modules, (C) design for multiple products, (D) modules with multifunctionality [Color figure can be viewed at wileyonlinelibrary.com]

conditions with detailed derivative information under the assumption of convexity. The application of such approaches to large-scale, nonconvex, and nonlinear process systems is challenging, which is mostly the case for PI systems. More advanced convexification techniques or surrogate-based methods for flexibility analysis are of importance such as those reviewed in Bhosekar and Ierapetritou.[88]

- Operability metrics that can simultaneously assess multiple operational aspects of such systems are very limited (e.g., flexibility, operability, safety, controllability, availability, reliability, maintenance). Some examples for this purpose include: the combined flexibility–reliability index to simultaneously cope with stochastic parameter variations and equipment failures proposed by Thomaidis and Pistikopoulos,[89] the integrated control-safety system design to account for operational safety via model predictive control proposed by Albalawi et al.,[90] and so on.
- One of the key characteristics of PI is to encourage novel technology development. The question thus arises on how to assess the practical operational performance of a brand new process with a lack of historical data (this question also exists for cost estimation and safety evaluation). In the cases when a process model is developed prior to experimental design, the above operability analysis approaches can help to provide a quantitative estimation of the operational performance at conceptual design stage based on available design and operational information, such as temperature/pressure, structural combination, process inventories, DOFs, and so on.

3.4 | Advanced model-based control with model reduction and dynamic optimization techniques

Extensive literature reviews have been provided on the control of PI and modular processes (e.g., reactive distillation, divided wall column, pressure swing adsorption).[11,41,91] Major research efforts in this area are oriented toward advanced model- and optimization-based control (particularly model predictive control, MPC) while conventional PID control is still widely applied in industrial practice.[41] MPC has been proved to provide superior control performance than PID control in a number of intensified/modular systems by accurately capturing their highly nonlinear behavior, complex dynamic characteristics, and strong variable interactions. Several case studies on MPC vs. PID control in intensified systems can be found in: PSA design and control optimization,[43] control of semi-continuous multicolumn countercurrent solvent gradient purification process,[62] and methyl tert-butyl ether reactive distillation.[92]

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However, the main technical challenges for MPC implementation in conventional and intensified chemical processes include[41]: (i) development and validation of rigorous dynamic models, (ii) computational complexity which hinders online MPC application, (iii) need for proper model reduction scheme, and (iv) realization of nonlinear MPC. In this context, the following technical needs are identified:

- A validated and standardized PI model library — which includes steady-state and dynamic process models as well as reduced order models and model reduction techniques.
- Model reduction techniques — to particularly support control and operational analysis. There have actually been extensive research work in the PSE community to apply surrogate modeling however mostly for (large scale) process synthesis, design and optimization leveraging machine-learning techniques.[88,93,94] For control and operational analysis, reliable model reduction techniques are needed to: (i) reduce the original highly complex and nonlinear process model to simplified linear/nonlinear model, which can be handled within MPC controller design while maintaining key dynamic characteristics and sufficient accuracy, and (ii) identification of process model from experimental or industrial dataset.
- Dynamic optimization algorithms — Existing dynamic optimization is limited with computational capacity and global optimization techniques. However, the implementation of online MPC requires to repeatedly solve the resulting nonlinear dynamic optimization problem (possibly also with repetitive model identification and reduction). Moreover, (mixed-integer) dynamic optimization algorithms are key enablers for simultaneous design and control optimization and real-time optimization — thus emphasizing the need of more efficient (global) dynamic optimization algorithms and solution tools.

An alternative to online MPC is the explicit/multi-parametric MPC, which can reduce the online computational load by reformulating and solving the MPC problem offline via multi-parametric programming.[95] Based on this, Pistikopoulos and co-workers proposed the PARametric Optimisation and Control (PAROC) framework, which is an integrated framework and software platform for the optimization and advanced model-based control of real-world process systems.[52] The proposed framework can also handle operational uncertainties on different time scales, that is, control, design, and scheduling.[80] The framework has also been extended to incorporate operability and safety metrics.[75]

3.5 | Integrated process design and operability/control

It has long been suggested to pay attention to operability, controllability, and safety under uncertainty during the early design phase rather than relying solely on posterior control systems and/or protection layers.[51,96] At each stage of conceptual design, different synthesis/design methods (e.g., superstructure-based synthesis, steady-state design optimization, dynamic high fidelity modeling) are adopted. Incorporating operability, safety, and control analysis in process synthesis stage helps to rapidly screen the large PI design space, as well as to deliver intensified designs with guaranteed operability performance. The match of information between design approaches and operational analysis methods is thus important to integrate these decisions as early as possible and to provide a consistent estimation of operational performance throughout the multiscale design procedure.

Table 3 gives an indicative list of integrated design and operability/control approaches applicable at each design stage. Operability analysis for the phenomena-based synthesis intensification is kind of challenging since the synthesis is done using abstract phenomena building blocks (e.g., reaction, separation) with no available equipment information (e.g., reactor, membrane). As an attempt for this purpose,

TABLE 3 Integrated multiscale design with operability/control analysis

	Design information	Operation information	Operability/control analysis
Phenomena based Synthesis and intensification	Structural layout of phenomena	Operating conditions	Flexibility analysis w/multiperiod design
	Pseudo sizing and cost estimation	(e.g., temperature and pressure)	Inherent safer design
	No equipment information available		(based on inventories and operating conditions)
Steady-state design and optimization	Equipment type selection	Operating conditions	Steady-state flexibility w/multiperiod design
	Equipment design parameters	Feasible/operable region	Operability-based design
	Equipment sizing		Structural controllability
	Operating and capital cost		Inherent safer design
Dynamic high fidelity modeling and simulation	Specific unit selection	Operating conditions	Dynamic flexibility w/multiperiod design
	Equipment (internal) design parameters	Open-loop dynamic response	Integrated design and control
	Equipment sizing	Controller design	(including simultaneous design and control)
	Operating and capital cost	Closed-loop dynamic behavior	Safety, reliability, resilience, and so on

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Tian and Pistikopoulos[74] applied flexibility analysis with multiperiod design to a methyl tert-butyl ether reactive separation case study using the phenomena-based GMF. The flexibility analysis was performed by analogizing each modular building block as a real mass or heat exchanger and then utilizing the flowsheet information on structural layout, stream flowrate, temperature/pressure, pseudo-sizing estimation, and so on. In this work, risk assessment for inherent safety considerations was also incorporated as model constraints resulting in an integrated GMF-flexibility-safety synthesis strategy. The risk assessment considered only the process hazard inventories and operating conditions to penalize large process design under extreme process conditions without detailed equipment considerations. It is worth highlighting that the consideration of flexibility and inherent safety even at the phenomena level can result in notable structural changes in the optimal process solutions — for example, the suggestion of using a reactive Petlyuk column instead of a reactive distillation column to ensure inherent safer design under feed flowrate uncertainties.[74]

Stepping to the equipment-based steady state and dynamic design, significant progresses have been made in the PSE community to explore design under uncertainty under multiple time scales. Detailed reviews can be found in Yuan et al.[97] and Burnak et al.,[82] while an indicative list is given as follows:

- Steady-state flexibility with multiperiod design — for example, heat exchanger network,[83] mass and heat exchange network[98]
- Dynamic flexibility with multiperiod design — for example, distillation column,[99] air separation system[100]
- Design with controllability considerations — for example, reaction-separation system,[101] divided wall column[102]
- Operability-based design — for example, membrane reactor[31,103]
- Integrated design and control — for example, combined heat and power system,[104] reactive distillation[92,105]

To identify the optimal PI/modular design given multiple operability criteria, specific metrics can be incorporated as inequality constraints in the process model to ensure that the resulting design can meet all the operational requirements. In addition, multiobjective optimization serves as a powerful tool to generate a number of Pareto optimal solutions with different design configurations and different levels of cost and operational performances. The shape of the Pareto curve can provide rich quantitative information on selecting a best solution from the trade-off solutions. For example, how much more cost to pay in exchange for a 10% increase of feasible operating region. The preferences of decision makers can also be taken into consideration to provide threshold limits which can constrain the decision space or to identify a utopia point which can be used to select the best trade-off solution via geometric optimization.

In addition to operability and control, sustainability is another essential factor to be explicitly considered at the early PI design stage. Tula et al.[37] developed a systematic framework and computer-aided software tool for sustainable process synthesis-intensification, which could generate intensified process solutions superior to base case designs in both environmental and economic aspects. Castillo-Landero et al.[106] proposed a process intensification methodology to minimize the number of pieces of equipment which incorporated economic, sustainability, and inherent safety metrics. A more recent work by Demirel et al.[107] combined the building block-based process synthesis intensification approach with multiobjective optimization to illustrate the trade-off between emissions and return on investment for different conventional and intensified process solutions. A number of excellent perspective papers and books have also been provided by the community on sustainable process design (e.g., Bakshi and Fiksel,[108] El-Halwagi,[109] and Pistikopoulos et al.[110])

3.6 | Framework for design, operability and control of PI and modular systems

In Section 2.4.3, we have highlighted the need for a systematic framework toward the full integration of process intensification, operation, and control which should be able to:

- Rapidly screen the combinatorial design space including both known conventional process solutions and "out-of-the-box" intensified ones
- Explore PI opportunities both via steady-state design (e.g., task integration, miniaturization) and dynamic operation (e.g., periodic operation)
- Evaluate the trade-off between scaling up and numbering up (i.e., centralized vs. modular design) with respect to economics, operability, control, and so on.
- Analyze the operability and control performances of PI and modular systems at conceptual design stage
- Systematically generate intensified and/or modular designs with guaranteed operability and control performances

As an attempt to address some of the above challenges, Tian et al.[75] introduced a step-wise framework to synthesize process intensified systems with guaranteed operability, safety, and control performances accounting for both steady-state design and dynamic operation. It has been showcased through a case study on the synthesis of intensified and operable reactive separation systems.[75] Ongoing research is extending the framework to encapsulate dynamic process intensification and modular systems design, as well as to develop more sophisticated approaches for multiscale PI design with operability considerations.

3.7 | Software tool for computer-aided design, operability analysis, and control optimization of PI and modular systems

A detailed review of available software tools for process operability, control, and safety can be found in the study by Tian et al.[11] However, these assessments are performed in "a posterior" manner after a process design is obtained. To the best of our knowledge, currently there is

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no available commercial software, based on which the operability and control of an intensified and modular process system can be ensured, or to integrate operational analysis with design and synthesis tasks. On a wider scope, there is also a lack of commercial software tools to: (i) generate novel conceptual process designs without prepostulation of equipment or flowsheet and accounts for process intensification opportunities, (ii) supply readily developed model library consisting of process models and physical property models on a consistent basis.

On the other hand, attempts have been initiated from academia to develop software prototypes for computer-aided process intensification. ProCAFD,[37] developed by Gani and co-workers as part of the ICAS tool set,[111] supports flowsheet synthesis and design using process-groups to represent unit operations and following a hierarchical procedure to generate optimal process alternatives with considerations of process safety, sustainability, and so on. More recently, the authors have published another toolkit, ProCACD,[112] which enables controller design and simultaneous design-control. MIPSYN[113] and Pyosyn[114] are two notable software prototypes supporting process synthesis. MIPSYN is one of the pioneers to introduce synthesis capabilities into an algebraic modeling environment with custom modeling — note that so far there is no available commercial software supporting process synthesis activities. As part of the Department of Energy IDAES project,[49] Pyosyn presents a newer synthesis framework using Generalized Disjunctive Programming model based in Python/Pyomo algebraic modeling platform. Another ongoing software prototype development, as part of the RAPID SYNOPSIS project,[115] aims to fully integrate the synthesis, design, operation, and control of intensified process configurations.

3.8 | Industrial perspectives

A first key question to the industry is how to decide on: (i) if to investigate in the research and implementation of PI and modular designs compared to conventional process designs, (ii) which new process solutions to be considered at an early design stage, and (iii) if the resulting process solutions can be profitable, operable, sustainable, and so on. To this end, the development of software (prototype) tools for computer-aided design and operational optimization of PI and modular systems, as discussed in Section 3.7, can be particularly beneficial. These tools should assist the decision makers to obtain a quantitative and holistic view of the plausible process solutions by: (i) analyzing the existing equipment or process to identify retrofit opportunities with process intensification methods and/or equipment; (ii) suggesting innovative process solutions which may be out-of-the-box of current practice; (iii) comparing intensified vs. conventional process solutions with respect to different performance metrics on cost, operability, sustainability, and so on; and (iv) generating multiple process solutions with different levels of cost/operability/sustainability performances for decision-making (e.g., via multiobjective optimization). To maximize the market competitivity, applicability and versatility of these new tools in industry, they should also be integrated with existing commercial software platforms (e.g., Aspen®, gPROMS®) to leverage the well-established modeling and simulation functionalities, physical databases, and model libraries, such as following the CAPE-OPEN Interface Standard. To facilitate the use of these tools by industrial professionals, the software (prototype) should be user-friendly with well-defined input and output information and user guides to provide engineering insights while avoiding overemphasis on modeling and optimization details. Training courses and workshops will also benefit the dissemination after the release of these academic-based tools.[116]

To further develop the suggested new modular process intensification technologies at a rapid pace, more detailed modeling and simulation platforms with advanced control are essential means to virtually test and improve the process.[47] Aglave et al.[27] emphasized the use of digitalization tools to create a closed-loop feedback between the digital twin (i.e., equipment or process models) and its physical counterpart (i.e., the unit or plant with real-time operating data) which enabled companies to continuously simulate, predict, and optimize the equipment and production systems throughout the life cycle of research and deployment.

Modularization opens up great opportunities toward more flexible, agile, and reliable industrial production systems. In addition to the intensified systems which are inherently modularized (e.g., membrane reactor, microreactor), having modular and parallel units (such as multiple reactive distillation columns) instead of one single unit can enhance the plant disturbance rejection performance and improve the operational flexibility to turn up/down production. However, in this case, the operability gains are at the price of additional capital and operating costs. This leads to the question on how to choose among a series of trade-off solutions, the decision of which should be made based on a detailed review of relevant regulations, business targets, expert opinions, and so on. Meanwhile, novel production and operation schemes should be further explored to achieve a win–win scenario for economic and operability performances leveraging modularization, for example, integrating multi-production and/or demand response-oriented process operation.

4 | CONCLUDING REMARKS

In this work, we have highlighted the operability and control challenges in process intensification and modular designs systems. It has also been discussed how to address these open questions built on recent advances on theoretical developments and computer-aided tools for process synthesis intensification, model-based operability analysis, advanced control, and operational optimization. While the discussions have been mostly based on individual PI equipment to identify the fundamental characteristics in such systems, further investigations are necessitated for integrated process flowsheet with multiple PI and modular designs. The use of multiple PI units may provide more promising process solutions in terms of energy savings, environmental impacts, and so on, while exhibiting more complex dynamic behaviors due to process integration.

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To conclude, we summarize the following key points and future perspectives:

- It is important to develop a formal theory and fundamental understanding of operability and control in PI and modular designs. On this basis, synergistic research developments should be explored with: (i) advances in novel design and synthesis approaches toward systematic modular process intensification, and (ii) advances in core PSE competencies on process operability analysis, model-based optimal control, and integrated design with operability/control.[110]
- A holistic framework and computer-aided software (prototype) platform for the design, operability analysis, and control optimization of PI and modular systems can benefit the development and decision-making for these novel technologies. A joint effort is necessitated from the community to: (i) develop a specialized PI model library and physical database (e.g., the ongoing RAPID projects on the Reaction Software Ecosystem[117] and the Physical Properties Database[118]), (ii) develop computational methods and algorithms, (iii) explore software integration and dissemination, and (iv) establish a collection of benchmarking case studies (e.g., the Tennessee Eastman process created by the Eastman Chemical Company).
- Hybrid mechanistic/data-driven modeling and simulation approaches can also play an important role to build process models with enhanced computational efficiency and/or incorporation of components both learned from data and first principles.[93,94]

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AUTHOR CONTRIBUTIONS

Efstratios Pistikopoulos: Conceptualization; methodology; supervision; writing-review & editing. **Yuhe Tian:** Formal analysis; methodology; writing-original draft. **Rahul Bindlish:** Conceptualization; supervision; writing-review & editing.

DATA AVAILABILITY STATEMENT

Data available on request from the authors

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REFERENCES

1. IEA. Tracking industry. https://www.iea.org/reports/tracking-industry-2019. 2019. Accessed June 24, 2020.
2. BP. BP Energy Outlook 2019. https://www.bp.com/en/global/corporate/energy-economics/energy-outlook.html. 2019. Accessed June 24, 2020.
3. Bielenberg J, Palou-Rivera I. The rapid manufacturing institute – reenergizing us efforts in process intensification and modular chemical processing. *Chem Eng Process*. 2019;138:49-54.
4. EFCE. European Roadmap for Process Intensification. https://efce.info. 2007. Accessed June 24, 2020.
5. Stankiewicz AI, Moulijn JA. Process intensification: transforming chemical engineering. *Chem Eng Prog*. 2000;96(1):22-34.
6. Van Gerven T, Stankiewicz A. Structure, energy, synergy, time – the fundamentals of process intensification. *Indus Eng Chem Res*. 2009;48(5):2465-2474.
7. Keil FJ. Process intensification. *Rev Chem Eng*. 2018;34(2):135-200.
8. Harmsen J. Process intensification in the petrochemicals industry: drivers and hurdles for commercial implementation. *Chem Eng Process*. 2010;49(1):70-73.
9. Harmsen GJ. Reactive distillation: the front-runner of industrial process intensification: a full review of commercial applications, research, scale-up design and operation. *Chem Eng Process*. 2007;46(9):774-780.
10. Yildirim Ö, Kiss AA, Kenig EY. Dividing wall columns in chemical process industry: a review on current activities. *Sep Purif Technol*. 2011;80(3):403-417.
11. Tian Y, Demirel SE, Hasan MF, Pistikopoulos EN. An overview of process systems engineering approaches for process intensification: state of the art. *Chem Eng Processing*. 2018;133:160-210.
12. Baldea M, Edgar TF, Stanley BL, Kiss AA. Modularization in chemical processing. *Chem Eng Prog*. 2018;114(3):46-54.
13. Weinfeld JA, Owens SA, Eldridge RB. Reactive dividing wall columns: a comprehensive review. *Chem Eng Process*. 2018;123:20-33.
14. Gude VG, Martinez-Guerra E. Green chemistry with process intensification for sustainable biodiesel production. *Environ Chem Lett*. 2018;16(2):327-341.
15. Stephen JL, Periyasamy B. Innovative developments in biofuels production from organic waste materials: a review. *Fuel*. 2018;214:623-633.
16. Coward T, Tribe H, Harvey AP. Opportunities for process intensification in the UKwater industry: a review. *J Water Process Eng*. 2018;21:116-126.
17. Suryawanshi PL, Gumfekar SP, Bhanvase BA, Sonawane SH, Pimplapure MS. A review on microreactors: reactor fabrication, design, and cutting-edge applications. *Chem Eng Sci*. 2018;189:431-448.
18. Baldea M, Edgar TF. Dynamic process intensification. *Curr Opin Chem Eng*. 2018;22:48-53.
19. Skiborowski M. Process synthesis and design methods for process intensification. *Curr Opin Chem Eng*. 2018;22:216-225.
20. Kiss AA, Jobson M, Gao X. Reactive distillation: stepping up to the next level of process intensification. *Indus Eng Chem Res*. 2018;58(15):5909-5918.
21. Daoutidis P, Allman A, Khatib S, et al. Distributed decision making for intensified process systems. *Curr Opin Chem Eng*. 2019;25:75-81.
22. Dias LS, Ierapetritou MG. Optimal operation and control of intensified processes – challenges and opportunities. *Curr Opin Chem Eng*. 2019;25:82-86.
23. Sitter S, Chen Q, Grossmann IE. An overview of process intensification methods. *Curr Opin Chem Eng*. 2019;25:87-94.
24. Masuku CM, Biegler LT. Recent advances in gas-to-liquids process intensification with emphasis on reactive distillation. *Curr Opin Chem Eng*. 2019;25:95-100.
25. Tian Y, Pistikopoulos EN. Synthesis of operable process intensification systems: advances and challenges. *Curr Opin Chem Eng*. 2019;25:101-107.
26. Demirel SE, Li J, Hasan MF. Systematic process intensification. *Curr Opin Chem Eng*. 2019;25:108-113.
27. Aglave R, Lusty J, Nixon J. Using simulation and digitalization for modular process intensification. *Chem Eng Prog*. 2019;115(3):45-49.
28. Jiang Z, Agrawal R. Process intensification in multicomponent distillation: a review of recent advancements. *Chem Eng Res Des*. 2019;147:122-145.

15475905, 2021, 5, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/aic.17204 by Texas A&M University Libraries, Wiley Online Library on [15/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

29. Stankiewicz AI, Yan P. 110th anniversary: the missing link unearthed: materials and process intensification. *Indus Eng Chem Res*. 2019;58(22):9212-9222.
30. Pereira CS, Patel BA. The role of process intensification in addressing the dual energy challenge. *Chem Eng Process*. 2019;142: 107545.
31. Gazzaneo V, Carrasco JC, Vinson DR, Lima FV. Process operability algorithms: past, present and future developments. *Indus Eng Chem Res*. 2019;59(6):2457-2470.
32. Tula AK, Eden MR, Gani R. Computer-aided process intensification: challenges trends and opportunities. *AIChE J*. 2020;66(1): e16819.
33. Adamu A, Russo-Abegão F, Boodhoo K. Process intensification technologies for CO2 capture and conversion – a review. *BMC Chem Eng*. 2020;2(1):1-18.
34. Baldea M. From process integration to process intensification. *Comput Chem Eng*. 2015;81:104-114.
35. Papalexandri KP, Pistikopoulos EN. Generalized modular representation framework for process synthesis. *AIChE J*. 1996;42(4):1010-1032.
36. Demirel SE, Li J, Hasan MF. Systematic process intensification using building blocks. *Comput Chem Eng*. 2017;105:2-38.
37. Tula AK, Babi DK, Bottlaender J, Eden MR, Gani R. A computer-aided software-tool for sustainable process synthesis-intensification. *Comput Chem Eng*. 2017;105:74-95.
38. Weber RS, Snowden-Swan LJ. The economics of numbering up a chemical process enterprise. *J Adv Manuf Process*. 2019;1(1–2):e10011.
39. Arora A, Li J, Zantye MS, Hasan MF. Design standardization of unit operations for reducing the capital intensity and cost of small-scale chemical processes. *AIChE J*. 2020;66(2):e16802.
40. Bhosekar A, Ierapetritou M. Modular design optimization using machine learning-based flexibility analysis. *J Process Control*. 2020;90:18-34.
41. Nikačević NM, Huesman AE, dPM HV, Stankiewicz AI. Opportunities and challenges for process control in process intensification. *Chem Eng Process*. 2012;52:1-15.
42. Yan L, Edgar TF, Baldea M. Dynamic process intensification of binary distillation based on output multiplicity. *AIChE J*. 2019;65(4):1162-1172.
43. Khajuria H, Pistikopoulos EN. Optimization and control of pressure swing adsorption processes under uncertainty. *AIChE J*. 2013;59(1): 120-131.
44. Becker T, Lier S, Werners B. Value of modular production concepts in future chemical industry production networks. *Eur J Operat Res*. 2019;276(3):957-970.
45. Allman A, Zhang Q. Dynamic location of modular manufacturing facilities with relocation of individual modules. *Eur J Operat Res*. 2020;286(2):494-507.
46. Pistikopoulos EN. Operability in process intensification and modular design. Foundations of Computer-Aided Process Design Conference 2019.
47. Bindlish R, Pistikopoulos EN. Operability challenges in process intensification: the role of dynamic modelling & advanced control. AIChE Southwest Process Technol Conference 2019.
48. Luyben M, Floudas C. Analyzing the interaction of design and control – 1. A multiobjective framework and application to binary distillation synthesis. *Comput Chem Eng*. 1994;18(10):933-969.
49. Miller DC, Siirola JD, Agarwal D, et al. Next generation multi-scale process systems engineering framework. *Comput Aided Chem Eng*. 2018;44:2209-2214.
50. Grossmann IE, Morari M. *Operability, Resiliency, and Flexibility: Process Design Objectives for a Changing World*. Proceedings of second international conference on foundations of computer-aided process design. Michigan: CACHE; 1983;931–1012.
51. Morari M. Effect of design on the controllability of chemical plants. *IFAC Proc Vol*. 1992;25(24):3-16.
52. Pistikopoulos EN, Diangelakis NA, Oberdieck R, Papathanasiou MM, Nascu I, Sun M. PAROC – an integrated framework and software platform for the optimisation and advanced model-based control of process systems. *Chem Eng Sci*. 2015;136:115-138.
53. Ye Y, Grossmann IE, Pinto JM, Ramaswamy S. Modeling for reliability optimization of system design and maintenance based on Markov chain theory. *Comput Chem Eng*. 2019;124:381-404.
54. Chin HH, Varbanov PS, Klemeš JJ, Benjamin MFD, Tan RR. Asset maintenance optimisation approaches in the chemical and process industries – a review. *Chem Eng Res Des*. 2020;164: 162-194.
55. Hasebe S. Design and operation of micro-chemical plants – bridging the gap between nano, micro and macro technologies. *Comput Chem Eng*. 2004;29(1):57-64.
56. Shao Y, Zavala VM. Modularity measures: Concepts, computation, and applications to manufacturing systems. *AIChE Journal*. 2020; 66(6). http://dx.doi.org/10.1002/aic.16965.
57. Christofides PD, Scattolini R, Pena DM, Liu J. Distributed model predictive control: a tutorial review and future research directions. *Comput Chem Eng*. 2013;51:21-41.
58. Wiles C, Watts P. Continuous process technology: a tool for sustainable production. *Green Chem*. 2014;16(1):55-62.
59. Lomel S, Falk L, Commenge J, Houzelot J, Ramdani K. The microreactor: a systematic and efficient tool for the transition from batch to continuous process? *Chem Eng Res Des*. 2006;84(5): 363-369.
60. Su Q, Nagy ZK, Rielly CD. Pharmaceutical crystallisation processes from batch to continuous operation using MSMPR stages: modelling, design, and control. *Chem Eng Process*. 2015;89:41-53.
61. Grossmann C, Ströhlein G, Morari M, Morbidelli M. Optimizing model predictive control of the chromatographic multi-column solvent gradient purification (MCSGP) process. *J Process Control*. 2010; 20(5):618-629.
62. Papathanasiou MM, Avraamidou S, Oberdieck R, et al. Advanced control strategies for the multicolumn countercurrent solvent gradient purification process. *AIChE J*. 2016;62(7):2341-2357.
63. Costandy JG, Edgar TF, Baldea M. Switching from batch to continuous reactors is a trajectory optimization problem. *Indus Eng Chem Res*. 2019;58(30):13718-13736.
64. Bîldea CS, Pătruţ{t\hskip-0.7ex\char "B8} C, Jørgensen SB, Abildskov J, Kiss AA. Cyclic distillation technology – a mini-review. *J Chem Technol Biotechnol*. 2016;91(5):1215-1223.
65. Biegler LT, Jiang L, Fox VG. Recent advances in simulation and optimal design of pressure swing adsorption systems. *Sep Purif Rev*. 2005;33(1):1-39.
66. Silveston PL, Hudgins RR. Periodic temperature forcing of catalytic reactions. *Chem Eng Sci*. 2004;59(19):4043-4053.
67. Sterman LE, Ydstie BE. The steady-state process with periodic perturbations. *Chem Eng Sci*. 1990;45(3):721-736.
68. Freund H, Sundmacher K. Towards a methodology for the systematic analysis and design of efficient chemical processes: part 1. From unit operations to elementary process functions. *Chem Eng Process*. 2008;47(12):2051-2060.
69. Peschel A, Freund H, Sundmacher K. Methodology for the design of optimal chemical reactors based on the concept of elementary process functions. *Indus Eng Chem Res*. 2010;49(21):10535-10548.
70. Feinberg M, Ellison P. General kinetic bounds on productivity and selectivity in reactor-separator systems of arbitrary design: principles. *Indus Eng Chem Res*. 2001;40(14):3181-3194.
71. Frumkin JA, Doherty MF. Target bounds on reaction selectivity via Feinberg's CFSTR equivalence principle. *AIChE J*. 2018;64(3):926-939.
72. Pérez-Cisneros ES, Sales-Cruz M. Thermodynamic analysis of the driving force approach: non-reactive systems. *Comput Chem Eng*. 2017;107:37-48.

15475905, 2021, 5, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/aic.17204 by Texas A&M University Libraries, Wiley Online Library on [15/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

73. Lopez-Arenas T, Sales-Cruz M, Gani R, Pérez-Cisneros ES. Thermodynamic analysis of the driving force approach: reactive systems. *Comput Chem Eng*. 2019;129:106509.
74. Tian Y, Pistikopoulos EN. Synthesis of operable process intensification systems – steady-state design with safety and operability considerations. *Indus Eng Chem Res*. 2018;58(15):6049-6068.
75. Tian Y, Pappas I, Burnak B, Katz J, Pistikopoulos EN. A systematic framework for the synthesis of operable process intensification systems – reactive separation systems. *Comput Chem Eng*. 2020;134:106675.
76. Cruz dFE, Manousiouthakis VI. Process intensification of reactive separator networks through the IDEAS conceptual framework. *Comput Chem Eng*. 2017;105:39-55.
77. Pattison RC, Touretzky CR, Johansson T, Harjunkoski I, Baldea M. Optimal process operations in fast-changing electricity markets: framework for scheduling with low-order dynamic models and an air separation application. *Indus Eng Chem Res*. 2016;55(16):4562-4584.
78. Cao Y, Swartz CL, Flores-Cerrillo J. Preemptive dynamic operation of cryogenic air separation units. *AIChE J*. 2017;63(9):3845-3859.
79. Dias LS, Pattison RC, Tsay C, Baldea M, Ierapetritou MG. A simulation-based optimization framework for integrating scheduling and model predictive control, and its application to air separation units. *Comput Chem Eng*. 2018;113:139-151.
80. Burnak B, Diangelakis NA, Katz J, Pistikopoulos EN. Integrated process design, scheduling, and control using multiparametric programming. *Comput Chem Eng*. 2019;125:164-184.
81. Otashu JI, Baldea M. Demand response-oriented dynamic modeling and operational optimization of membrane-based chlor-alkali plants. *Comput Chem Eng*. 2019;121:396-408.
82. Burnak B, Diangelakis NA, Pistikopoulos EN. Towards the grand unification of process design, scheduling, and control – Utopia or reality? *Processes*. 2019;7(7):461.
83. Halemane KP, Grossmann IE. Optimal process design under uncertainty. *AIChE J*. 1983;29(3):425-433.
84. Grossmann IE, Floudas CA. Active constraint strategy for flexibility analysis in chemical processes. *Comput Chem Eng*. 1987;11(6):675-693.
85. Dimitriadis VD, Pistikopoulos EN. Flexibility analysis of dynamic systems. *Indus Eng Chem Res*. 1995;34(12):4451-4462.
86. Lin CT. Structural controllability. *IEEE Trans Autom Control*. 1974;19(3):201-208.
87. Georgakis C, Uztürk D, Subramanian S, Vinson DR. On the operability of continuous processes. *Control Eng Pract*. 2003;11(8):859-869.
88. Bhosekar A, Ierapetritou M. Advances in surrogate based modeling, feasibility analysis, and optimization: a review. *Comput Chem Eng*. 2018;108:250-267.
89. Thomaidis TV, Pistikopoulos E. Optimal design of flexible and reliable process systems. *IEEE Trans Reliability*. 1995;44(2):243-250.
90. Albalawi F, Durand H, Christofides PD. Process operational safety via model predictive control: recent results and future research directions. *Comput Chem Eng*. 2018;114:171-190.
91. Kiss AA, Bildea CS. A control perspective on process intensification in dividing-wall columns. *Chem Eng Process*. 2011;50(3):281-292.
92. Tian Y, Pappas I, Burnak B, Katz J, Pistikopoulos EN. Simultaneous design & control of a reactive distillation system – a parametric optimization & control approach. *Chem Eng Sci*. 2021;230:116232.
93. Kim SH, Boukouvala F. Machine learning-based surrogate modeling for data-driven optimization: a comparison of subset selection for regression techniques. *Optim Lett*. 2019;14:989-1010.
94. Williams B, Cremaschi S. Surrogate model selection for design space approximation and surrogate-based optimization. *Comput. Aided Chem. Eng*. Proceedings of the 9th International Conference on Foundations of Computer-Aided Process Design. Netherlands: Elsevier; 2019:353-358.
95. Bemporad A, Morari M, Dua V, Pistikopoulos EN. The explicit linear quadratic regulator for constrained systems. *Automatica*. 2002;38(1):3-20.
96. Pistikopoulos E. Uncertainty in process design and operations. *Comput Chem Eng*. 1995;19:553-563.
97. Yuan Z, Chen B, Sin G, Gani R. State-of-the-art and progress in the optimization-based simultaneous design and control for chemical processes. *AIChE J*. 2012;58(6):1640-1659.
98. Papalexandri K, Pistikopoulos E. A multiperiod MINLP model for the synthesis of flexible heat and mass exchange networks. *Comput Chem Eng*. 1994;18(11–12):1125-1139.
99. Mohideen MJ, Perkins JD, Pistikopoulos EN. Optimal design of dynamic systems under uncertainty. *AIChE J*. 1996;42(8):2251-2272.
100. Zhu Y, Legg S, Laird CD. Optimal design of cryogenic air separation columns under uncertainty. *Comput Chem Eng*. 2010;34(9):1377-1384.
101. Fisher WR, Doherty MF, Douglas JM. The interface between design and control. 1. Process controllability. *Indus Eng Chem Res*. 1988;27(4):597-605.
102. Serra M, Perrier M, Espuna A, Puigjaner L. Study of the divided wall column controllability: influence of design and operation. *Comput Chem Eng*. 2000;24(2–7):901-907.
103. Gazzaneo V, Lima FV. Multilayer operability framework for process design, intensification, and modularization of nonlinear energy systems. *Indus Eng Chem Res*. 2019;58(15):6069-6079.
104. Diangelakis NA, Burnak B, Katz J, Pistikopoulos EN. Process design and control optimization: a simultaneous approach by multiparametric programming. *AIChE J*. 2017;63(11):4827-4846.
105. Bernal DE, Carrillo-Diaz C, Gomez JM, Ricardez-Sandoval LA. Simultaneous design and control of catalytic distillation columns using comprehensive rigorous dynamic models. *Indus Eng Chem Res*. 2018;57(7):2587-2608.
106. Castillo-Landero A, Ortiz-Espinoza AP, Jimenez-Gutierrez A. A process intensification methodology including economic, sustainability, and safety considerations. *Indus Eng Chem Res*. 2018;58(15):6080-6092.
107. Demirel SE, Li J, El-Halwagi M, Hasan MF. Sustainable process intensification using building blocks. *ACS Sustain Chem Eng*. 2020;8:17664-17679.
108. Bakshi BR, Fiksel J. The quest for sustainability: challenges for process systems engineering. *AICHE J*. 2003;49(6):1350-1358.
109. El-Halwagi MM. *Sustainable Design Through Process Integration: Fundamentals and Applications to Industrial Pollution Prevention, Resource Conservation, and Profitability Enhancement*. Amsterdam: Elsevier Inc.; 2017.
110. Pistikopoulos EN, Barbosa-Povoa A, Lee JH, et al. Process systems engineering – the generation next? *Comput. Chem. Eng*. 2021;107252. https://doi.org/10.1016/j.compchemeng.2021.107252
111. Gani R, Hytoft G, Jaksland C, Jensen AK. An integrated computer aided system for integrated design of chemical processes. *Comput Chem Eng*. 1997;21(10):1135-1146.
112. Tula AK, Wang J, Chen X, Mansouri SS, Gani R. ProCACD: a computer-aided versatile tool for process control. *Comput Chem Eng*. 2020;136:106771.
113. Kravanja Z, Grossmann IE. PROSYN – an MINLP process synthesizer. *Comput Chem Eng*. 1990;14(12):1363-1378.
114. Chen Q, Kale S, Bates J, et al. Pyosyn: a collaborative ecosystem for process design advancement. 2019 AIChE annual meeting. 2019.
115. Pistikopoulos EN, Hasan MMF, Kwon JS, et al. SYNOPSIS – Synthesis of Operable Process Intensification Systems. RAPID Institute Project 9.3, DE-EE0007888-09-03. 2020.
116. Hasan MMF, Pistikopoulos EN, El-Halwagi M, Young W. COMPLETE – Computer-aided, Model-based Process Intensification Learning, Training, and Education. RAPID Institute Project 3.2.5.
117. Vlachos DG. RAPID reaction software ecosystem. RAPID Institute Project 9.5. 2020.

15475905, 2021, 5, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/aic.17204 by Texas A&M University Libraries, Wiley Online Library on [15/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

118. Sholl D. An experimentally verified physical properties database for sorbent selection and simulation. RAPID Institute Project 9.6. 2020.

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